Sports Impact Study: Do New Sports Venues Raise Local House Prices? Evidence from U.S. Metro-Level Housing Data

Quan Nguyen & Michael Zhang

ECON 4200: Senior Seminar

Dr. John Connaughton

December 1, 2025

I. Introduction

Professional sports have not only become a popular and patriotic pastime in the United States, but the development of the sports industry has made important economic contributions, especially in metropolitan areas that host these sports teams and venues. Nowadays, as cities compete with one another to attract and/or retain major league sports teams, the cost of new sports stadiums and arenas has increased, with construction costs heading into the billions this very decade. Because these large investments are oftentimes funded with public money, it is important for us to understand if these new sports venues benefit the metro's community at all.

If a new venue raises home values, then it may justify pouring public money devoted to it. On the other hand, if there are little to no benefits, then taxpaying residents may be supporting these initiatives for little to no return. For these reasons, it is important that we have clear evidence on whether these investments benefit the public, specifically, whether they benefit home values for residents.

To examine these effects, we use a difference-in-differences (DiD) framework, supported by an event study to estimate how new sports venue openings affect metro-level home values. We combine monthly Zillow Home Value Index data for every U.S. metropolitan area from 2000 to 2025 with a comprehensive dataset of stadium construction from Bradbury, Coates, and Humphreys (2025). The DiD approach allows us to compare treated and untreated metros, while the event study traces the timing and gradual buildup of housing-market changes around the opening date.

Our results show that new sports venues produce modest but consistent increases in home values, with an average post-opening effect of about 3.9%. These gains develop gradually within one to three years rather than appearing immediately. We also find that higher cost projects generate larger increases: approximately 0.21% per $100 million in construction spending, while public versus private financing shares of venue investment do not significantly change the effect once cost is already controlled for.

Overall, this study provides updated national evidence on how stadium openings shape metro-level housing markets, offering insight into whether the considerable public spending on these projects translates into real economic gains for local communities.

II. Literature review

We look at 8 similar studies on how opening new sports venues affect housing prices in an area over time. The differences across these studies usually depend on local context, how the stadium is financed, and the methods used to estimate the impact. These studies, summarized below, help establish expectations for how venue-related shocks may influence the housing market. We examine studies that point towards negative, mixed, and positive results.

On the negative side, Dehring, Depken, and Ward (2006) examined the Dallas–Fort Worth area using an event study framework and found that when a publicly funded NFL stadium was announced in Arlington, residential property values decreased by roughly 1.5%. This decline occurred before construction began, indicating that homeowners reacted negatively to the anticipated tax burden associated with the project. Their study demonstrates how the financial structure of stadium deals can influence property markets even in the absence of physical changes to the surrounding neighborhood.

On the mixed side, Coates and Humphreys (2012) analyzed census block groups within five miles of major U.S. sports facilities using a hedonic pricing model with spatial controls, and found that home values were higher near stadiums in some cities but lower or insignificant in others. Propheter (2023) reached a similar conclusion while studying the Golden State Warriors' relocation. Using a relocation-based comparison of housing markets near both the old and new sites, the study found rising prices around the new arena and declining prices near the old one. The direction of the effect depended entirely on whether the neighborhood gained or lost the stadium, which makes the overall impact mixed rather than purely positive or negative.

On the other hand, several studies saw positive venue effects. Firstly, Ahlfeldt and Maennig (2010) analyzed Berlin's multifunctional arenas using a hedonic pricing model and found that land values increased within a three-kilometer radius of the stadiums, with the strongest gains appearing in neighborhoods that were undergoing redevelopment. Secondly, Chikish, Humphreys, and Nowak (2019) also identified a positive impact in Oklahoma City, where property values rose following the opening of a new arena and the arrival of a full time NBA franchise. Finally, Funahashi and Cardazzi (2025), by using a DiD

approach and examining 27 stadium projects across Japan, found that stadium openings raise nearby property values.

Our contribution differs from the existing literature in three ways. First, while some prior studies focus on individual cities or a small set of venues, we use a large metro-month panel covering 377 U.S. metros from 2000 to 2025 and include all major league sports venue openings since 2000. Second, instead of examining home values in particular neighborhoods, blocks, or distance bands around a venue opening, we study the effect on metro-wide home prices using Zillow's Home Value Index, a highly precise estimate of housing prices that is robust to seasonality. Third, we include heterogeneity models that test whether higher construction cost and public financing generate different responses to home prices. Ultimately, by combining DiD and event study models, we can both assess the impact of a venue opening on a metro's home prices and see how those effects evolve over time.

III. Data Description

From Zillow, we measure housing market prices using Zillow's Home Value Index (ZHVI), a monthly index that tracks the value of a typical home in each U.S. metro. Zillow calculates ZHVI by first using an advanced neural network model that automatically appraises each individual property based on recent sales of similar homes, property characteristics and features, location, public records, and listing data. These appraisals are called *Zestimates,* which is Zillow's predicted sale price for a home. After removing outliers, Zillow then aggregates these *Zestimates* across all homes in a metro and adjusts for seasonality to construct the ZHVI, which provides a clean and accurate measure of underlying home values in each U.S. metro. The data runs from 2000-2025 and includes both metropolitan areas (MSAs) and micropolitan areas (MiSAs). Because we only care about the impact of sports venues on metropolitan areas, we remove all micropolitan areas from our analysis.

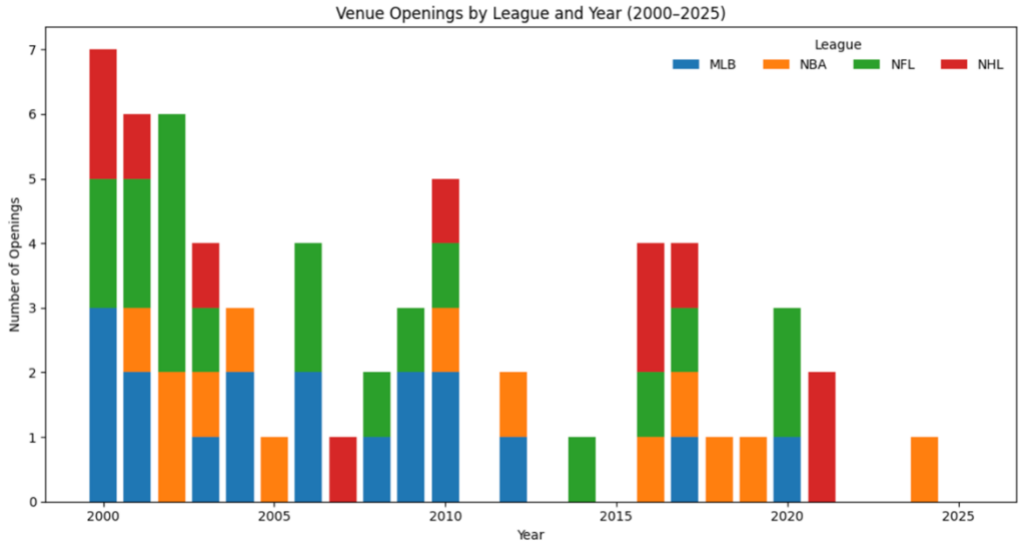
From the *US Major-League Sports Stadium and Arena Construction Costs (1909–2027)* updated dataset (Bradbury, Coates, and Humphreys, 2023), we observe not only when and where venues open for all four major U.S. sports leagues (MLB, NBA, NFL, and NHL), but

also their total construction costs and public financing amounts, expressed in 2020 millions of dollars. We then match each venue to its corresponding U.S. metropolitan area in the Zillow dataset and merge the two datasets together to form a 25-year metro-month panel, which we use to study how new venues impact housing prices.

Very few metros receive more than one venue opening in the years spanning 2000-2025, but for those that do (Las Vegas, New York, Dallas, etc.), we keep only their earliest opening venues to avoid overlapping post-treatment periods. Although we lose some later, potentially impactful openings in a few large metros, having duplicate shocks to the same metros would complicate the interpretation of our DiD estimates. Additionally, some metro-months have missing ZHVI data, so we simply drop observations before running any regressions. The resulting combined dataset is an unbalanced panel that tracks home prices and opening information for 377 metros across up to 309 months, yielding 105,824 metro-month total observations.

IV. Summary Statistics

Having described the raw data and the construction of the metro-month panel, we now present several summary statistics that illustrate the structure of the sample. Figure 1 below plots venue opening by league and year, ranging from 2000 to 2025.



Venue openings aren't super common: the U.S. only sees 61 openings from 2000-2025, with 42 occurring in 2000-2010 alone. This indicates that openings are staggered and sparse, with only a small subset of metros seeing a new venue in our 25-year window.

Figure 2 below contains boxplots comparing the total construction cost and public financing cost among arenas and stadiums. "Stadiums" are used to describe the larger venues that host NFL and MLB teams, and "arenas" reflect the smaller venues that are occupied by NHL and NBA teams. Stadiums are generally costlier, with a median cost of a stadium is $627 Million (in 2020$), compared to an arena's $425 Millon. This difference is simply because stadiums are larger, which makes them generally more expensive to build. The gap in costs motivate our cost-interaction specification that we test later in the paper.

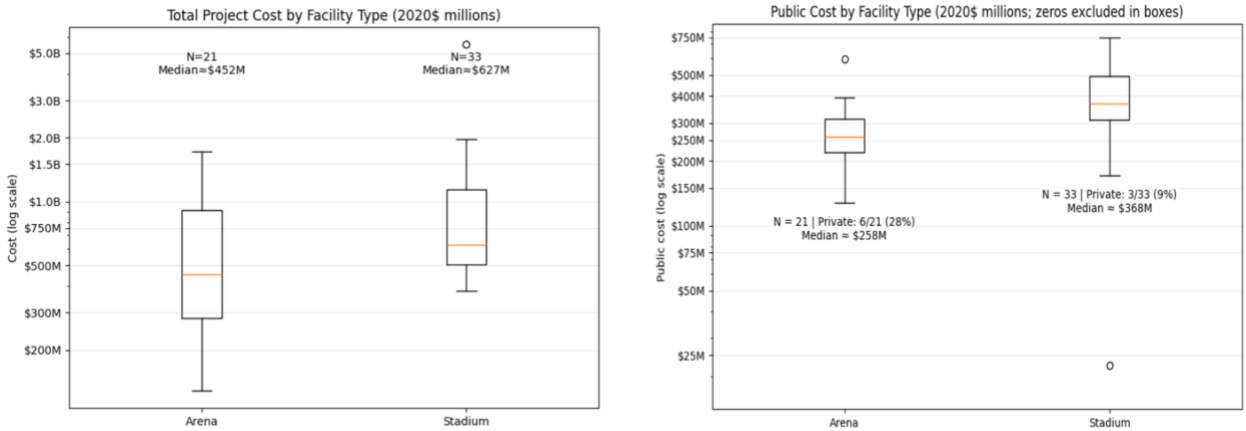
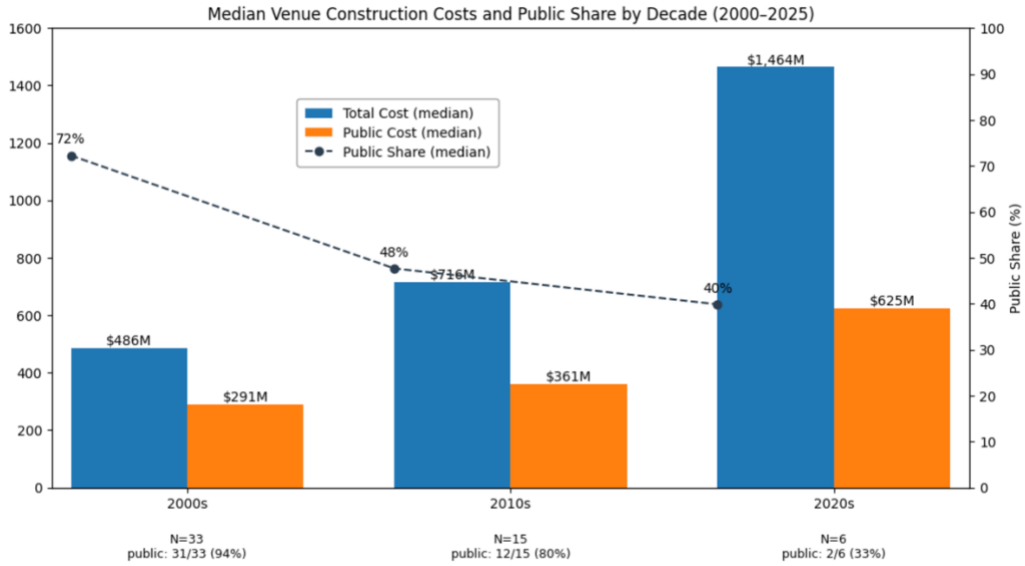
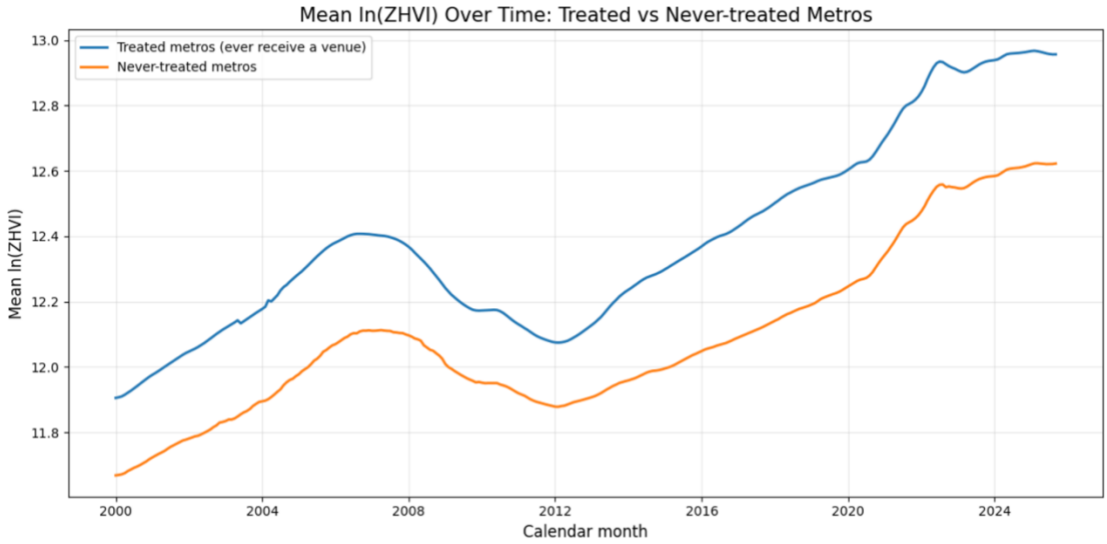


Figure 3 below, which was inspired by Bradbury, Coates and Humphreys (2023), summarizes the median total cost, median public cost, and median public share by decade. Construction costs have increased dramatically over time, likely due to the increase in the demand for larger and more luxurious facilities with top of the notch technology. Although public financing costs move in the same direction as construction costs, the share of public financing has decreased over time, consistent with an increase in the number of privately funded venues. These trends highlight the importance of understanding whether these larger investments, often with publicly backed dollars, generate local economic benefits.



Finally, figure 4 plots the mean log ZHVI over time for metros that ever receive a new venue and those that never do. Both groups follow a similar price trajectory over time, with the treated metros always a level greater than the untreated, which is unsurprising since larger, wealthier cities are more likely to gain new sports venues. Furthermore, the roughly parallel movement of the two series provides preliminary support for the parallel-trends assumption that is the backbone of our difference-in-differences and event-study models.



V. Methodology

This section explains the econometric tools that we use to estimate and understand how major league sports venue openings affect metro-level home prices. We start with a baseline staggered difference-in-differences model, then we add interactions to see how effects vary with project cost and public financing. Finally, we conduct an event-study to look at pre-trends and examine the dynamic response of home prices after a venue opening.

A. Baseline Difference-in-Differences Model

For each metro *i,* we denote t_0 as the month of its earliest venue opening between 2000 and 2025. We define our main independent variable, $Treated_i \cdot Post_{it}$, as an interaction that equals 1 for metro-months in the post-opening period for treated metros and 0 for not-yet-treated or never-treated metros. Our dependent variable, $lnZHVI_{it}$, is the log Zillow Home Value Index for metro *i* in month *t*. Because venue openings are staggered across metros, we include metro fixed effects, a_i , and month fixed effects, ω_t. Metro fixed effects remove time-invariant differences across metros (e.g. long-run prices, local labor markets, etc.), while month fixed effects remove nationwide shocks in each month (e.g. national economic swings, policy changes, etc.). By including these fixed effects, identification comes from movements in a treated metro's home prices, relative to the untreated metros. Thus, we estimate the staggered difference-in-differences model via two-way fixed effects:

$$lnZHVI_{it} = \beta_1(Treated_i \cdot Post_{it}) + a_i + \omega_t + \epsilon_{i,t}$$

Here, β_1 captures whether treated metros, in the months after opening a venue, experience higher home prices than to untreated metros. More specifically, β_1 is the average difference-in-differences in log home prices between treated and untreated metros, after vs. before the venue opening, controlling for metro and month fixed effects.

B. Cost-Interacted Model

As shown earlier, the median construction cost for venues has increased drastically since the 2000s as fans and teams demand bigger and better venues. Because venue

projects vary so greatly in scale, we next examine whether the treatment effect depends on the size of the investment by interacting it with $totalcost_i$, the total construction cost of a venue in metro *i*, measured in 2020 millions of dollars. Formally, we estimate:

$$lnZHVI_{it} = \beta_1(Treated_i \cdot Post_{it}) + \beta_2(Treated_i \cdot Post_{it} \cdot totalcost_i) + a_i + \omega_t + \epsilon_{i,t}$$

Here, β_2 measures how the post opening effect changes with project scale. More specifically, β_2 captures how much larger the post-opening change in home prices is for a metro whose venue costs one additional unit of $totalcost_i$. Additionally, β_1 now represents the post opening effect for a hypothetical venue with zero construction cost, which is a meaningless baseline and hence not interpretable.

C. Public-Financing Interacted Model

Privately funded venues have become a bit more popular in recent times, with four out of the nine privately funded venues since 2000 coming from the 2020s alone. To examine whether the source of funding affects housing prices, we interact treatment indicator with $publiccost_i$, which measures the portion of the venue's $totalcost_i$ that is publicly funded.

$$lnZHVI_{it} = \beta_1 Post_{it} + \beta_2(Treated_i \cdot Post_{it} \cdot totalcost_i) + \beta_3(Treated_i \cdot Post_{it} \cdot publiccost_i) + a_i + \omega_t + \epsilon_{i,t}$$

Here, β_3 tests whether publicly funded projects generate meaningfully different effects than private ones, and β_2 still measures how the post opening impact on home prices changes with project scale. Again, β_1 is not interpretable because it is the post opening effect for a hypothetical venue with zero construction cost and zero public funding.

D. Monthly Event Study Model

A limitation of the two way-fixed effects difference-in-differences models is that it averages all the post opening months into one coefficient, so we can't see when prices move or check for parallel pre-trends. Additionally, because the treatment timing is staggered, two-way fixed effects can be biased if already-treated metros are used as controls for later-treated metros. Moreover, earlier-treated metros contribute more heavily to the estimated

average effect because they have more post-months, making it harder to reliably interpret our coefficients. For these reasons, we complement our two-way fixed effects model with an event-study, which allows us to examine the dynamic responses of housing prices and check for parallel trends. Formally, we estimate:

$$lnZHVI_{it} = a_i + \omega_t + \sum_{k \neq -1} \theta_k * 1\{k = t - t_0\} + \epsilon_{i,t}$$

Here, we still use metro fixed effects a_i and month fixed effects ω_t to strip out fixed differences across metros and nationwide cycles and shocks. The term, $k = t - t_0$ is the event time, which counts the number of months relative to the opening month t_0, and $1\{k = t - t_0\}$ is the indicator that metro *i* is *k* months away from its opening in month *t.* We omit *k* = -1, the month right before the opening, and use it as the reference period. Thus, each θ_k measures the effect on home values *k* months from the opening month. More specifically, θ_k is the average difference in log home prices *k* months before and after opening, relative to the month right before opening, after controlling for metro and month fixed effects.

VI. Results

In all regressions, we report two-way clustered standard errors by metro and month. Clustering the standard errors by metro allows the error terms in a metro to be correlated over time, while clustering by month allows the errors to be correlated across metros in the same month.

A. Baseline Differences-in-Differences Estimate

$$lnZHVI_{it} = \beta_1(Treated_i \cdot Post_{it}) + a_i + \omega_t + \epsilon_{i,t}$$

The table below represents the results from our baseline staggered differences-in-differences model. The coefficient on our treatment indicator, β_1 = 0.039, infers that after a metro's venue opens, its Zillow Home Value Index is about 3.9% higher on average than it would have been without the opening, relative to metros that haven't received a new venue in the same months, controlling for metro and month fixed effects. The p-value of 0.0585 indicates that the estimate is statistically significant at the 10% level. The within R^2 implies

that only 0.4% of the within-metro, over-time variation in $lnZHVI$ is explained by the treatment indicator variable. Overall, the result from this regression gives some initial evidence that major-league sports venues in the U.S. generate decent gains in metro-wide housing prices. However, the causality of the difference-in-differences estimate here depends on the parallel-trends assumption, which we look at later in the event study.

Dep. Variable:	ln_zhvi	R-squared:	0.0012
Estimator:	PanelOLS	R-squared (Between):	0.0004
No. Observations:	105824	R-squared (Within):	0.0041
Date:	Fri, Nov 21 2025	R-squared (Overall):	0.0004
Time:	19:47:17	Log-likelihood	8.41e+04
Cov. Estimator:	Clustered		

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
post	0.0388	0.0205	1.8921	0.0585	-0.0014	0.0789

B. Cost-Interacted Estimates

$$lnZHVI_{it} = \beta_1(Treated_i \cdot Post_{it}) + \beta_2(Treated_i \cdot Post_{it} \cdot totalcost_i) + a_i + \omega_t + \epsilon_{i,t}$$

The table below shows the results from interacting our treatment indicator with total construction cost. The coefficient on the main treatment variable, β_1 = 0.029, is not statistically significant at any conventional level, but it is not very interpretable since it represents the effect for a hypothetical venue with zero construction cost.

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:                 ln_zhvi   R-squared:                        0.0017
Estimator:                    PanelOLS   R-squared (Between):              0.0003
No. Observations:               105824   R-squared (Within):               0.0043
Date:                 Fri, Nov 21 2025   R-squared (Overall):              0.0004
Time:                         19:47:19   Log-likelihood                 8.413e+04
Cov. Estimator:              Clustered
```

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
post	0.0229	0.0230	0.9989	0.3178	-0.0221	0.0680
post_x_totalcost	2.141e-05	7.806e-06	2.7432	0.0061	6.114e-06	3.672e-05

The main coefficient of interest is the interaction term's coefficient, β_2. Here, β_2 = $2.14 \cdot 10^{-5}$, and is statistically significant (p-value = 0.006). Essentially, what this means is that home prices increase by $2.14 \cdot 10^{-5}$ for every additional $1 million in project cost. By scaling it by 100, we can interpret β_2 as: for each additional $100 million in project construction costs, measured in 2020 dollars, the post-opening effect on home prices increases by around 0.21%. For example, a $700 million venue would be associated with an additional 1.5% percentage point increase. The results here highlights that larger investments generate slightly larger metro-wide housing prices.

C. Public-Financing Interacted Estimates

$$lnZHVI_{it} = \beta_1 Post_{it} + \beta_2(Treated_i \cdot Post_{it} \cdot totalcost_i) + \beta_3(Treated_i \cdot Post_{it} \cdot publiccost_i) + a_i + \omega_t + \epsilon_{i,t}$$

In addition, we extend the model to examine whether the source of financing (public or private) affects housing market responses at all. The interaction between the treatment indicator and total construction cost remains statistically significant (p-value = 0.039), with each additional $100 million in project cost resulting in a 0.19% increase in home value. Now, the interaction between the treatment indicator and $publiccost$ (β_3 = -

0.016) is small and insignificant (p-value = 0.637). These results, as shown below, suggest that the results from publicly financed venues aren't different than privately financed ones, after controlling for overall venue investment, $totalcost_i$.

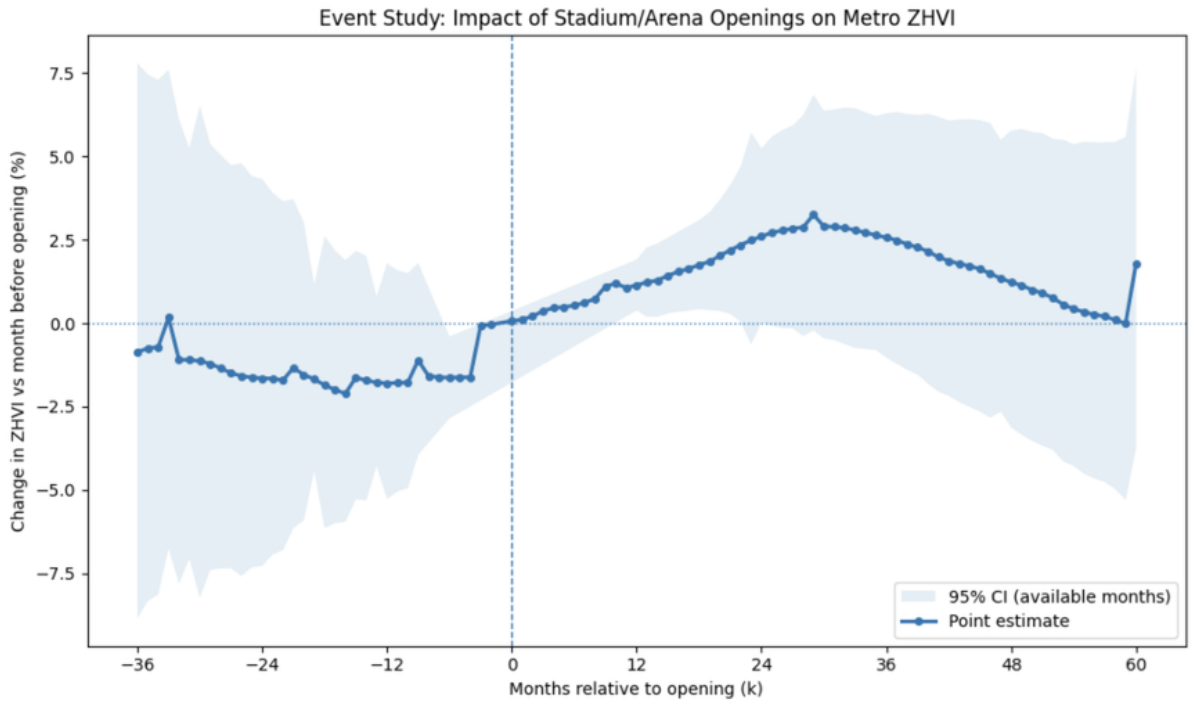
```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:                 ln_zhvi   R-squared:                       0.0017
Estimator:                    PanelOLS   R-squared (Between):             0.0003
No. Observations:               105824   R-squared (Within):              0.0043
Date:                 Fri, Nov 21 2025   R-squared (Overall):             0.0004
Time:                         19:47:21   Log-likelihood                 8.413e+04
Cov. Estimator:              Clustered

                               Parameter Estimates
================================================================================
                  Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
--------------------------------------------------------------------------------
post                 0.0388     0.0328     1.1819     0.2372     -0.0255      0.1031
post_x_totalcost  1.873e-05  9.052e-06     2.0687     0.0386   9.844e-07   3.647e-05
post_x_publicAny    -0.0162     0.0344    -0.4717     0.6371     -0.0836      0.0511
================================================================================
```

D. Monthly Event Study Estimates

$$lnZHVI_{it} = a_i + \omega_t + \sum_{k \neq -1} \theta_k * 1\{k = t - t_0\} + \epsilon_{i,t}$$

The final figure below represents the monthly event-study estimates, where each $\widehat{\theta_k}$ is a regression coefficient. For example, $\widehat{\theta_k}$ = 0.032 corresponds to a 3.2% higher home value for treated metros, relative to the month before opening. For the pre-period months before opening (k < 0), it is evident that nothing dramatic was happening to home values before venue openings. If anything, housing prices were lower in the entire pre period relative to the month before opening. In the opening month (k = 0), $\widehat{\theta_0} = 0.0006$ or 0.06%, implying that there is not much of an initial jump and that housing markets don't react instantly. However, in the post-opening months (k = 1 to 30), home prices rise gradually, reaching a $\widehat{\theta_{30}} = 2.8\%$ increase in home values 30 months after opening. For the later

period months (k = 30 to 60), the long-run effect on home values flattens, but wider confidence intervals points to uncertainty, and we can't say for sure what the long-run effects are since they are not precisely estimated.



VI. Conclusions

This paper examined whether new major-league sports venues create noticeable changes in metro-level housing values in the United States. Our results show that venue openings are followed by modest but meaningful increases in metro-level home values, averaging around 3.9%. These changes do not happen immediately; instead, values tend to rise gradually over one to three years. We also find that higher-cost projects lead to larger effects, while the mix of public and private financing does not appear to change the outcome once total cost is considered.

Although the home value gains from the findings are not incredibly large, they do suggest that new venues could positively contribute to regional housing markets. Also, the effects are limited and fade in the long run, which means they should not be viewed as the core reason for approving public subsidies.

This study has its limits. Metro-level ZHVI data may miss neighborhood-level differences, and we do not examine distributional effects. Future work could study transaction-level data or explore variation across financing structures and team types. Since population differs greatly between metros, future work could also measure effects in per-capita terms. The within R^2 is quite low, which suggests we could've used better fixed effects, as suggested by Dr. John Connaughton. Finally, it could be the case that venue timing is endogenous to housing markets as cities might build when values are already rising, which could inspire the use of an instrumental variable for better identification.

Overall, our results show that stadiums can raise housing values, but the impact is gradual and moderate, offering a more grounded view of what these projects bring to urban areas.

VII. References

Ahlfeldt, G.M. & Maennig, W. 2010, 'Impact of sports arenas on land values: Evidence from Berlin', The Annals of Regional Science, vol. 44, no. 2, pp. 205–227.

Bradbury, J.C., Coates, D. & Humphreys, B.R. 2025, US Major-League Sports Stadium and Arena Construction Costs (1909–2026), research dataset, Bagwell Center for the Study of Markets and Economic Opportunity, Kennesaw State University.

Chikish, Y., Humphreys, B.R. & Nowak, A. 2019, 'Sports facilities and residential property values: Evidence from the construction of new arenas and stadiums', Regional Science and Urban Economics, vol. 77, pp. 42–59.

Coates, D. & Humphreys, B.R. 2012, 'Game Attendance and Competitive Balance in Major League Baseball', Journal of Sports Economics, 13(1), pp. 20–43.

Coates, D. & Humphreys, B.R. 2008, 'Do economists reach a conclusion on subsidies for sports franchises, stadiums, and mega-events?', Econ Journal Watch, vol. 5, no. 3, pp. 294–315.

Cobb County Communications 2022, Braves stadium information, Cobb County Government, Marietta, GA.

Dehring, C.A., Depken, C.A. & Ward, M.R. 2007, 'The impact of stadium announcements on residential property values: Evidence from a natural experiment in Dallas–Fort Worth', Contemporary Economic Policy, vol. 25, no. 4, pp. 627–638.

Funahashi, H. & Cardazzi, A. 2024, 'An examination of the externalities of neighborhood stadiums on housing land values', paper presented at the Japan Society of Sports Industry 33rd Annual Conference, Japan.

Propheter, G. 2023, 'Sports facilities as a housing amenity: Do prices follow facilities?', Journal of Sports Economics, vol. 24, no. 4, pp. 443–474.